

		pressure $p = \frac{1}{3} \times 2.282 \times 10^{-5} \times 1900^2 = 27.456 \text{ Pa}$
12	C	equating $n = \frac{pV}{RT}$ at 40 m deep and at the surface, we get $\frac{(\rho gh + p_{atm}) \times 20 \text{ cm}^3}{R(4 + 273.15)} = \frac{p_{atm} \times V}{R(20 + 273.15)}, \text{ and}$ $\rho gh = 1200 \times 9.81 \times 40 = 4.709 \times 10^5 \text{ Pa}$ So $V = \left(\frac{4.709 + 1.0}{1.0} \right) \left(\frac{293.15}{277.15} \right) \times 20 \text{ cm}^3 = 120.8 = 120 \text{ cm}^3$
13	B	$a = g - r\omega^2$